Predictive Modeling of Student Dropout Risk:

A Phased Approach for Proactive Intervention

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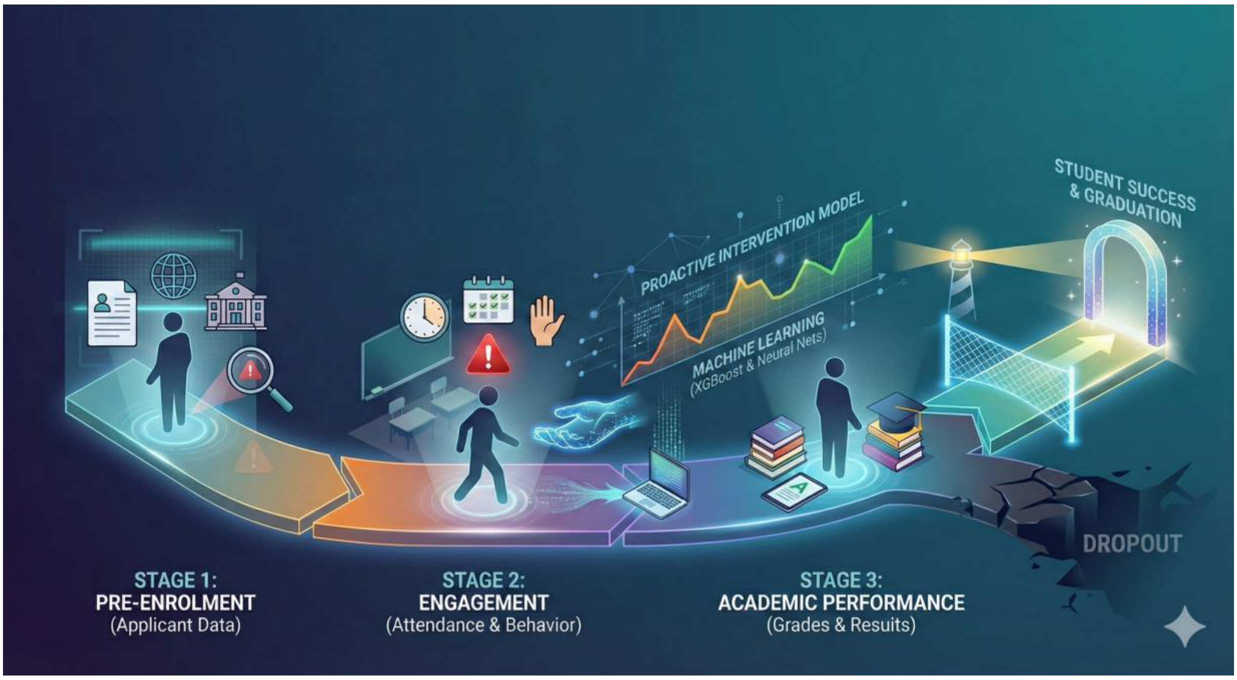

Project Summary

This report presents a three-stage machine learning framework to predict student dropout risk using application, engagement, and academic performance data.

- The goal is to help Study Group identify at risk students early and target support where it will have the most impact.

How to Use This Report

Use the findings and visuals that follow to guide when to intervene, which students to prioritize, and how to design a tiered retention strategy across the three stages.

Proactive Intervention to Support Student Success

1.0 Introduction

Student retention is a core strategic priority for Study Group. Dropout erodes revenue, weakens institutional reputation, and most importantly reflects an interrupted academic journey.
The objective is to build supervised learning models that predict dropout risk at multiple points in the student lifecycle to enable proactive, targeted interventions.

To mirror how data becomes available, the analysis is structured into three stages:

- **Stage 1**: Applicant and course information available pre-enrolment, enabling an initial risk score before a student starts.
- **Stage 2**: Stage 1 data plus attendance, capturing early engagement behaviour.
- **Stage 3**: Stage 2 data plus academic performance, incorporating direct evidence of progress through module results.

2.0 Methods

Staged Data Framework

The three-stage framework follows the sequence of data capture. Separate models are trained at each stage to show who is at risk and how early this can be known, which guides the timings and intensity of interventions.

Data Preparation and Modelling Strategy

Key steps:

- **Class Imbalance:** Around 15% of students are dropouts and 85% completers. To avoid models defaulting to the majority class, class weighting (*e.g., scale_pos_weight*) was applied so that dropout cases are penalized more heavily when misclassified.
- **Data Encoding:** Categorical variables such as nationality, coursename, and centrename were one-hot encoded so that tree-based models and neural networks can use them.
- **Model Selection:** Two algorithms were used: **XGBoost** (a high-performance gradient-boosting algorithm for tabular data), and a **Neural Network**. Both were first run as baselines and subsequently tuned through hyperparameter optimization.
- **Evaluation Metrics:** Accuracy, Precision, Recall, and AUC were reported. Because the goal is to find at risk students, recall for the dropout class is the main KPI and thresholds were set to favor higher recall over precision.

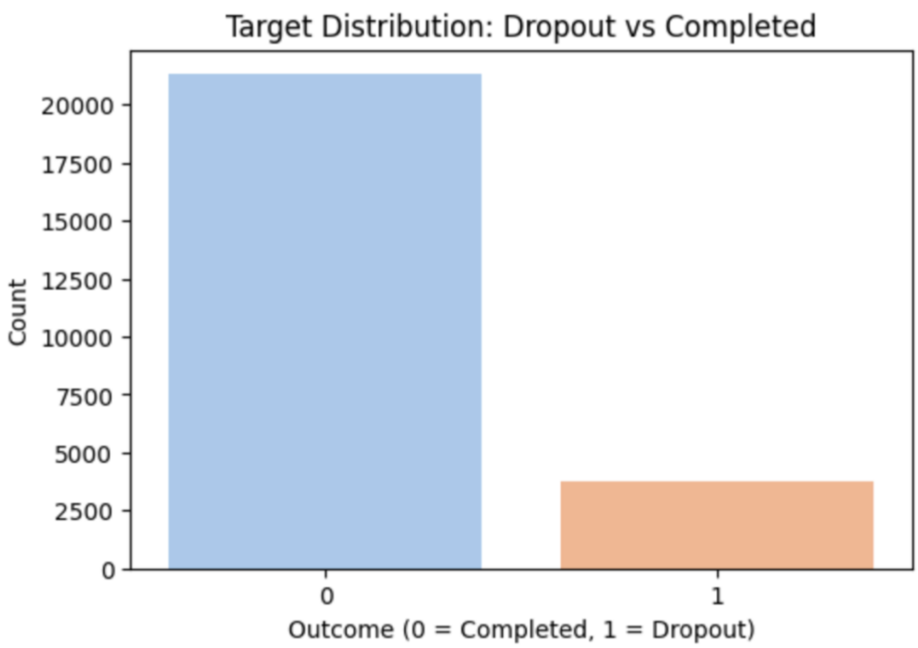

Figure 1- Class Imbalance

3.0 Results: Phased Model Performance Analysis

3.1 Stage 1 – Pre-Enrolment Risk Assessment (Applicant Data)

Using only the initial applicant and course data available, before a student's first day, the models were able to establish a solid baseline for risk assessment. The analysis identified key demographic and institutional factors correlated with dropout.

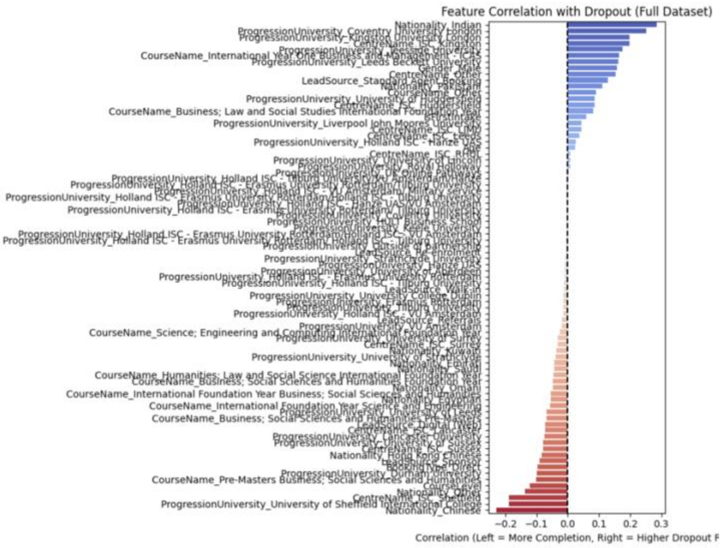

Figure 2 - Correlation Analysis with Dropout – STAGE 1

- **Primary Risk Drivers**: Progression University, nationality (e.g., Chinese, Hong Kong, Indian) Centre, and lead source (e.g., Standard Agent Booking) are dominant features. At this point, the model is effectively using demographic and institutional proxies for preparedness and support environment.
- **Tuned XGBoost**: The model achieved an overall Accuracy 0.85, AUC 0.8784. Crucially, it demonstrated a recall of 0.77 for the dropout class, capturing over three-quarters of potential dropouts before enrolment.
- **Recall-Optimised Neural Network:** The final tuned Neural Network model correctly identified 74% of all dropouts (recall) while maintaining a high overall accuracy of 87%. This result was achieved by adjusting the classification threshold to 0.65, which successfully shifted the model's focus towards identifying dropouts, making it a valuable tool for enabling proactive, pre-enrolment outreach.

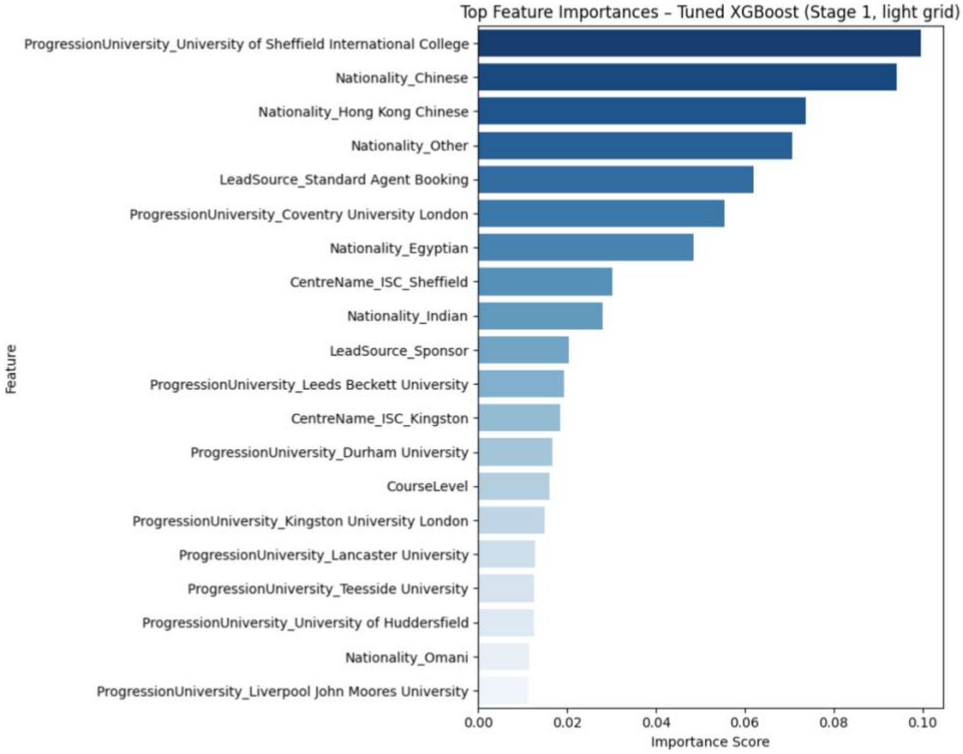

Figure 3 - XGBoost Top Feature Importance - STAGE 1

3.2 Stage 2 – Adding Engagement (Attendance)

When attendance information is added, both models improve. The table below summarizes the uplift from Stage 1 to Stage 2 (baseline models).

Table 1- Stage 1 vs Stage 2 (Baseline Models)

Model	Metric	Stage 1: Application Only	Stage 2: App + Engagement
XGBoost (baseline)			
	Accuracy	0.84	0.85
	AUC	0.88	0.90
	Dropout Recall	0.77	0.82
Neural Network (Base)			
	Accuracy	0.89	0.90
	AUC	0.87	0.90
	Dropout Recall	0.56	0.60

The inclusion of attendance data provides a genuine uplift in predictive power for both models. The XGBoost model benefited more significantly in its baseline form, with its dropout recall increasing by five percentage points.

This stage also marks a shift in feature importance. While segment-level factors like intended university and nationality remain critical, UnauthorisedAbsenceCount emerges as a key predictive feature. This demonstrates that poor attendance acts as a powerful early warning sign, amplifying the risk signals that were already present at the application stage.

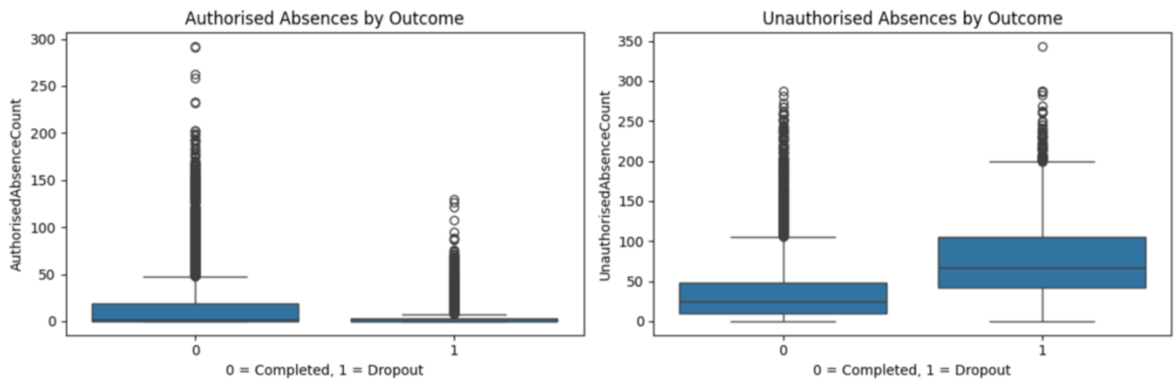

Figure 4 - Engagement Behaviour; Completers vs. Dropouts

3.3 Stage 3 – Adding Academic Performance

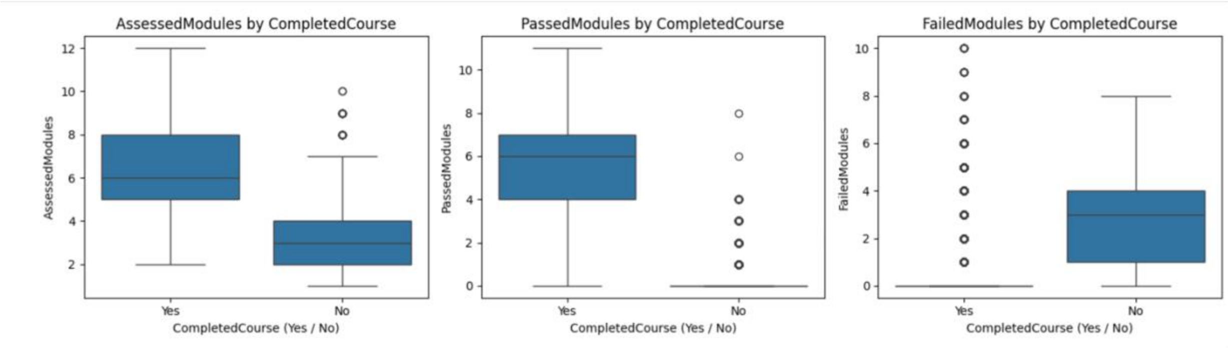

Figure 5 - Assessment Behaviour; Completers vs. Dropouts

When academic performance is added, the picture becomes clear. Completers have more modules assessed and passed and almost no failed modules, while dropouts cluster around fewer assessed and passed modules with a long tail of failures. The correlation chart reinforces this: FailedModules and UnauthorisedAbsenceCount have the strongest positive correlation with dropout.

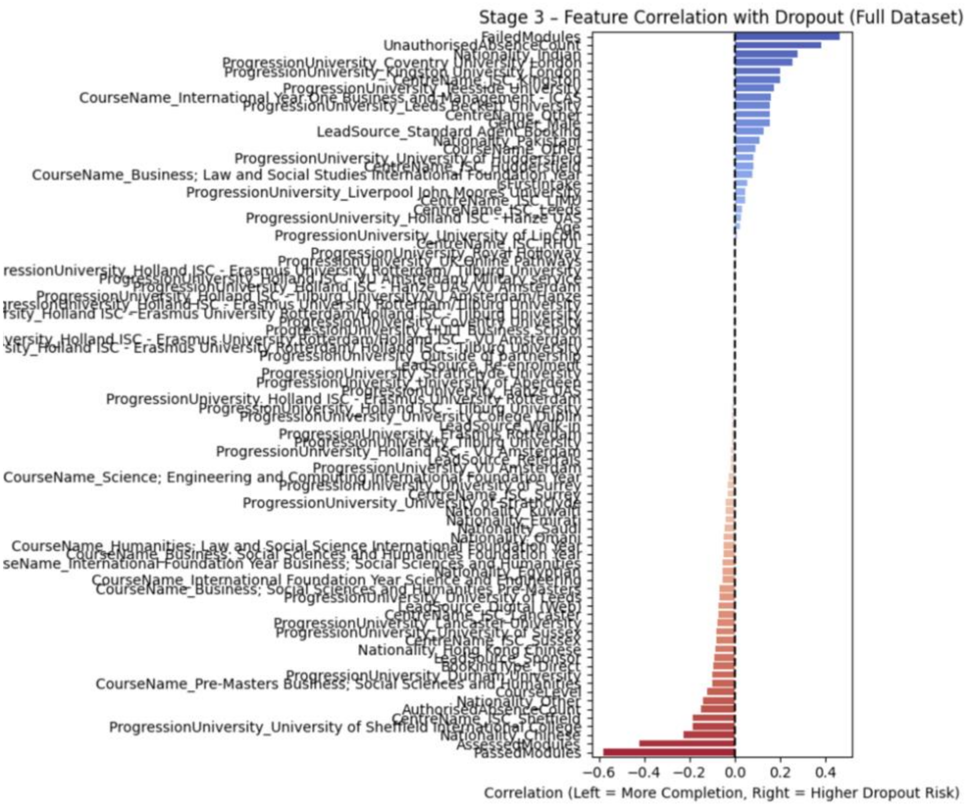

Figure 6 - Feature Correlation with Dropouts - Stage 3

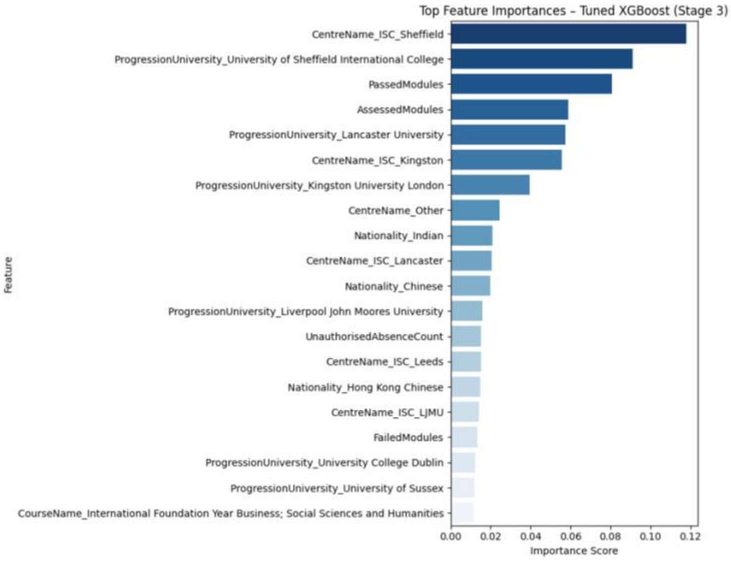

Figure 7 - XGBoost Feature Importance - STAGE 3

The tuned XGBoost feature importance plot confirms that performance is now central to prediction. CentreName_ISC_Sheffield and the progression route to the University of Sheffield International College remain important, but PassedModules and AssessedModules are among the top drivers of model output. Early demographic variables still matter, but they are clearly subordinate to direct evidence of academic struggle or success.

Table 2 - Stage 2 vs. Stage 3 (Baseline Models)

Model	**Metric**	**Stage 2: App + Engagement**	**Stage 3: App + Eng + Performance**
XGBoost (baseline)			
	Accuracy	0.85	0.93
	AUC	0.90	0.99
	Dropout Recall	0.82	0.94
	Dropout Precision	0.49	0.71
Neural Network (Base)			
	Accuracy	0.90	0.96
	AUC	0.90	0.98
	Dropout Recall	0.60	0.82
	Dropout Precision	0.70	0.92

The metrics track what the visuals signal. Once academic data is introduced, both models move from inference based on proxies to confirming risk using hard performance evidence. AUC approaches 1.0, and both recall and precision for the dropout class increase sharply.

- **Tuned XGBoost (Stage 3):** Accuracy 0.97, AUC 0.9934, dropout recall 0.93, precision 0.85.
- **Recall optimised Neural Network (Stage 3):** Accuracy 0.96, AUC 0.9840, with 0.88 recall and 0.84 precision at a 0.70 threshold.

In practical terms, once the first set of grades is in, Study Group can identify high risk students with high confidence and route them into intensive support with minimal noise.

4.0 Conclusion and Recommendations

This project shows that dropout risk follows a predictable evolution that supports a phased intervention strategy. By aligning predictive modelling with the student lifecycle—from enrolment to engagement and finally academic performance—we found that predictive power increases significantly as richer data becomes available.

Key Insights & Strategic Value

- **Evolving Risk Indicators:** The basis of risk shifts from demographic proxies (Stage 1) to behavioural warning signs like attendance (Stage 2), and finally to definitive academic evidence (Stage 3).
- **Model Reliability:** By Stage 3, both XGBoost and Neural Network models achieve over **96% accuracy**, confirming their readiness for operational deployment.
- **Strategic Tuning:** The models are optimized for high **Recall**. We deliberately accept lower precision (more "false alarms") to drastically reduce missed dropouts, ensuring the widest possible safety net for at-risk students.

Recommendations for Implementation

To maximize retention, Study Group should deploy a **Tiered Intervention System** aligned with these data stages:

1. **Stage 1 (Pre-Enrolment):** Use broad, low-cost communications to build early connections with higher-risk cohorts.
2. **Stage 2 (Engagement):** Trigger automated alerts and personal check-ins when students exhibit unauthorized absences, catching disengagement before it escalates.
3. **Stage 3 (Performance):** Utilize academic results to flag students for urgent, intensive, and personalized support from tutors and welfare staff.
4. **Operational Focus:** Prioritize resources for systemic hotspots identified by the models, such as specific centres (e.g., ISC Sheffield) and high-risk pathways.